NanoGT Match Report: Fabry disease

Disease: Fabry disease (ORPHA:324)
Primary gene: GLA
Gene CDS: 1290 bp
Inheritance: X-linked dominant
Target tissues scored: liver, kidney, heart, CNS

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: DISEASE HETEROGENEITY: Fabry disease has classic and late-onset phenotypes with different tissue involvement. CNS/cardiac delivery requirements vary by phenotype.; Multi-system disease; define a primary therapeutic target tissue before selecting route/vector; Vector does not naturally cover all annotated disease tissues: cns, heart, kidney, liver.

Disease Mechanism Evidence

Molecular mechanism: loss of function
Mechanistic detail: Alpha-galactosidase A lysosomal enzyme deficiency
Gene-addition compatibility: compatible
Preferred modality class: gene addition or cross correction
Evidence level/status: direct / source_linked_needs_review
Evidence summary: Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction
Evidence source: OMIM GLA gene entry

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint readiness: liver/metabolic targets may have biochemical biomarkers, but biomarker correction must be linked to clinical benefit.
- Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
- Endpoint risk: muscle/cardiac diseases may need functional, respiratory, imaging, or cardiac endpoints that progress slowly and vary by age/stage.
- Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	8.9/10	High	approved
2	ST-920	AAV2/6	8.8/10	High	phase1/2
3	RGX-121	AAV9	8.7/10	High	phase3
4	Hemgenix	AAV5	8.5/10	High	approved

Rank	Program	Vector	Score	Confidence	Approval
5	Roctavian	AAV5	8.5/10	High	approved

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 8.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.86**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1290bp / cargo 8000bp (16% utilized)
- Precedent target match: cns
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- LOF inheritance — compatible for gene replacement
- Pathway match: leukodystrophy
- Disease mechanism: loss of function — Alpha-galactosidase A lysosomal enzyme deficiency
- Gene-addition modality compatibility: supports gene addition

- Mechanism evidence: Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction
- Mechanism source: OMIM GLA gene entry (https://omim.org/entry/300644)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GLA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- DISEASE HETEROGENEITY: Fabry disease has classic and late-onset phenotypes with different tissue involvement. CNS/cardiac delivery requirements vary by phenotype.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, heart, kidney, liver
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #2: ST-920

Precedent disease: Fabry disease
Vector: AAV2/6
Tissue target: liver
Composite score: 8.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.76**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1290bp / cargo 4700bp (27% utilized)
- Vector tropism plus precedent target match: liver
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- Inheritance match (X-linked dominant <-> XL)
- Pathway match: lysosomal_storage
- Disease mechanism: loss of function — Alpha-galactosidase A lysosomal enzyme deficiency
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction
- Mechanism source: OMIM GLA gene entry (https://omim.org/entry/300644)
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GLA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- DISEASE HETEROGENEITY: Fabry disease has classic and late-onset phenotypes with different tissue involvement. CNS/cardiac delivery requirements vary by phenotype.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, heart, kidney
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: RGX-121

Precedent disease: Mucopolysaccharidosis type II
Vector: AAV9
Tissue target: CNS/liver
Composite score: 8.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.67**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1290bp / cargo 4700bp (27% utilized)
- Vector tropism plus precedent target match: cns, liver
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- Inheritance match (X-linked dominant <-> XL)
- Pathway match: lysosomal_storage
- Disease mechanism: loss of function — Alpha-galactosidase A lysosomal enzyme deficiency
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction
- Mechanism source: OMIM GLA gene entry (https://omim.org/entry/300644)

- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GLA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- DISEASE HETEROGENEITY: Fabry disease has classic and late-onset phenotypes with different tissue involvement. CNS/cardiac delivery requirements vary by phenotype.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: kidney
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Hemgenix

Precedent disease: Hemophilia B
Vector: AAV5
Tissue target: liver
Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.52**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1290bp / cargo 4700bp (27% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked dominant <-> XL)
- Different pathway (lysosomal_storage vs coagulation)
- Disease mechanism: loss of function — Alpha-galactosidase A lysosomal enzyme deficiency
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction
- Mechanism source: OMIM GLA gene entry (https://omim.org/entry/300644)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GLA standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- DISEASE HETEROGENEITY: Fabry disease has classic and late-onset phenotypes with different tissue involvement. CNS/cardiac delivery requirements vary by phenotype.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: heart, kidney
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Roctavian

Precedent disease: Hemophilia A
Vector: AAV5
Tissue target: liver
Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.52**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1290bp / cargo 4700bp (27% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked dominant <-> XL)
- Different pathway (lysosomal_storage vs coagulation)
- Disease mechanism: loss of function — Alpha-galactosidase A lysosomal enzyme deficiency
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: Fabry is caused by deficient GLA enzyme activity; secreted enzyme strategies can support systemic cross-correction
- Mechanism source: OMIM GLA gene entry (https://omim.org/entry/300644)

- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
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